$$\sum_j \mathbf{M}|j\rangle\alpha_j = \sum_j \beta_j|j\rangle.$$

The last step is to take the inner product of both sides with a particular basis vector $\langle k|$, resulting in

$$\sum_j \langle k|\mathbf{M}|j\rangle\alpha_j = \sum_j \beta_j\langle k|j\rangle. \qquad (3.1)$$

To make sense of this result, remember that $\langle k|j\rangle$ is zero if j and k are not equal, and 1 if they are equal. That means that the sum on the right side collapses to a single term, β_k.

On the left side, we see a set of quantities $\langle k|\mathbf{M}|j\rangle\alpha_j$. We can abbreviate $\langle k|\mathbf{M}|j\rangle$ with the symbol m_{kj}. Notice that each m_{kj} is just a complex number. To see why, think of **M** operating on $|j\rangle$ to give some new ket-vector. The inner product of $\langle k|$ with this new ket-vector must be a complex number. The quantities m_{kj} are called the *matrix elements* of **M** and are often arranged into a square $N \times N$ matrix. For example, if $N = 3$, we can write the symbolic equation

$$\mathbf{M} = \begin{pmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{pmatrix}. \qquad (3.2)$$

This equation involves a slight abuse of notation that would give a purist indigestion. The left side is an abstract linear operator and the right side is a concrete representation of it in a particular basis. Equating them is sloppy but it should not cause confusion.

Now let's revisit Eq. 3.1 and replace $\langle k|\mathbf{M}|j\rangle$ with m_{kj}. We get